

CSA0716-Computer Networks

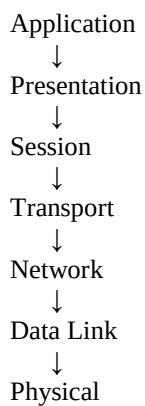
CO1-AT2

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1. Identify the Missing Layers

Given Concept Map



Missing Layers

- **Session Layer**
- **Data Link Layer**

Explanation

Session Layer

- Establishes, manages, and terminates communication sessions.
- Provides synchronization and recovery during data transfer.

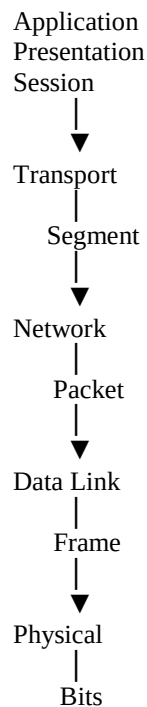
Data Link Layer

- Converts packets into frames.
- Performs framing, error detection, and node-to-node delivery using MAC addresses.

Importance for Reliable Communication

- The Session Layer ensures uninterrupted communication through synchronization.
- The Data Link Layer detects transmission errors and delivers frames reliably between adjacent devices.

2. Concept Map: Data Units and OSI Layers



Analysis

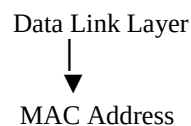
OSI Layer		Data Unit	Function
Transport	Segment	Divides data into segments	
Network	Packet	Adds IP address and routes data	
Data Link	Frame	Adds MAC address and error detection	
Physical	Bits	Converts data into electrical/optical/radio signals	

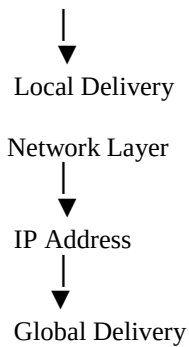
Conclusion

As data moves downward through the OSI layers, headers are added (encapsulation). At the receiver, headers are removed (decapsulation), ensuring successful communication.

3. MAC Address and IP Address Concept Map

Concept Map





Explanation

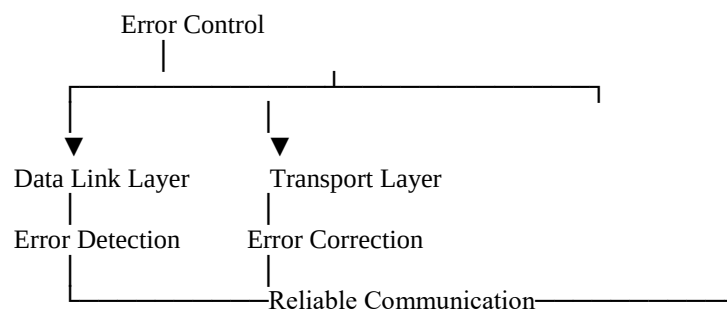
- **MAC Address**
 - Used for communication within the same local network (LAN).
 - Ensures correct device-to-device delivery.
- **IP Address**
 - Used for communication across different networks.
 - Enables routers to forward packets to the correct destination.

Analysis

The Network Layer delivers packets to the correct network using IP addresses, while the Data Link Layer delivers frames to the correct device using MAC addresses. Together, they ensure complete end-to-end communication.

4. Error Control Concept Map

Concept Map



Analysis

Data Link Layer

- Detects transmission errors using CRC or checksum.
- Requests retransmission between directly connected devices.

Transport Layer

- Performs end-to-end error recovery.

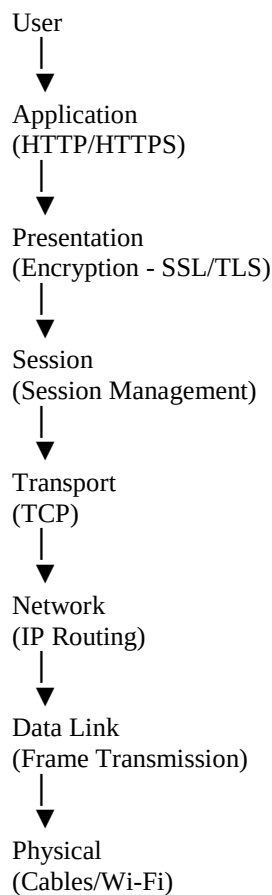
- Uses acknowledgements (ACKs), sequence numbers, and retransmissions.

Conclusion

The Data Link Layer detects errors locally, while the Transport Layer ensures complete error-free delivery between source and destination.

5. Concept Map for Web Browsing

Concept Map



Analysis

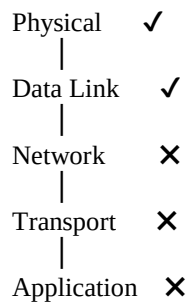
Layer Failure	Effect
Physical	No signal transmission
Data Link	Frame errors and communication failure
Network	Packets cannot reach destination
Transport	Data loss and connection failure
Session	Session disconnected
Presentation	Encryption/decryption fails
Application	Web page cannot load

Conclusion

All OSI layers work together. Failure in one layer interrupts the entire communication process.

6. Network Troubleshooting Concept Map

Given Concept Map



Analysis

- Physical Layer is working correctly.
- Data Link Layer is functioning properly.
- Failure begins at the **Network Layer**.

Possible Causes

- Incorrect IP address
- Router failure
- Incorrect subnet mask
- Gateway configuration error
- Routing problem

Since the Network Layer fails:

- Transport Layer cannot establish TCP/UDP communication.
- Application Layer cannot access network services.

Troubleshooting Steps

1. Check IP address configuration.
2. Verify default gateway.
3. Test connectivity using **ping**.
4. Verify router status.
5. Check routing tables.
6. Trace the path using **tracert**.

Conclusion

The concept map helps identify that the problem starts at the **Network Layer**, making troubleshooting faster and more systematic.